

GVPT Data Science Camp

Introduction and Setup

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University of Maryland

Today's Plan

- ▶ Welcome and introductions

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- ▶ What is R, and why do we use it?

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- ▶ Installing R and RStudio
- ▶ A tour of RStudio
- ▶ AI, LLMs, and agentic coding

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- ▶ Gentle introduction to the many tools you will use to conduct and present research.

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- ▶ Please ask questions about anything.

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- ▶ Fun fact: I used to operate a college basketball burner Twitter account

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- ▶ Data exists in structured and unstructured places
- ▶ A PhD in political science is a versatile degree

- ▶ Getting to know your cohort

Learning Objectives

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- ▶ Introduction to R and RStudio

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- ▶ Getting to know your cohort
- ▶ Introduction to R and RStudio
- ▶ Brief discussion of AI, LLMs, and agentic coding

R and R Studio

What is R and why do we use it?

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- ▶ R is a free, open-source programming language designed for statistical computing, data analysis, and data visualization
- ▶ Over the past four decades, quantitative research has dominated social science, and R provides us the ability to conduct such research

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- ▶ Follow the setup prompts; keep the default settings. Click “Finish” when complete

What is RStudio?

- ▶ RStudio is an **integrated development environment (IDE)** we use to write code

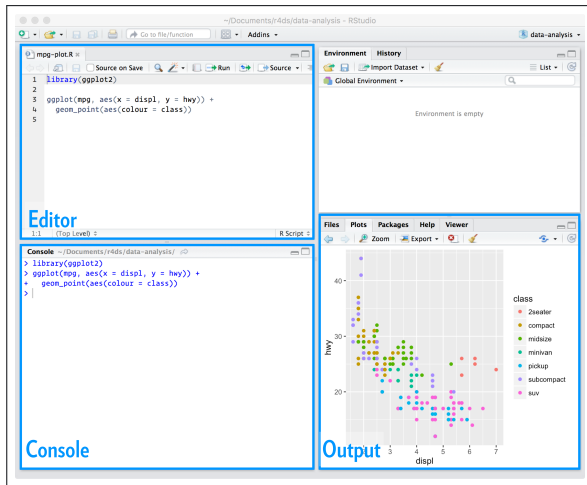
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- ▶ Run the setup:
 - ▷ Windows: open the downloaded **.exe** file and follow the setup prompts
 - ▷ Mac: open the downloaded **.dmg** disk image and drag the RStudio icon into your Applications folder

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- ▶ Click “Finish” or exit the installer once the process is complete

A Tour of RStudio



Source: *R for Data Science* (R4DS)

AI, LLMs, and Agentic Coding

- ▶ AI (Artificial Intelligence) is a branch of computer science that develops systems to do tasks that normally require human intelligence

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- ▶ LLMs (Large Language Models) are a type of AI that are trained on massive datasets to produce human-like text
- ▶ Agentic coding is a software development approach where autonomous AI agents plan, write, and test code with minimal human intervention

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- ▶ Set up R and RStudio
- ▶ Brief introduction to the exciting world of AI

- ▶ Data Visualization (Tuesday)

Plan for the Rest of the Week

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- ▶ Data Manipulation (Wednesday)

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- ▶ Data Presentation (Thursday)

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- ▶ Student Presentations (Thursday)
- ▶ AI in Data Science (Friday)

Questions?